ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.6CP Structural Checklist for Building Type S3: Metal Building Frames

Table 17-12. Collapse Prevention Structural Checklist for Building Type S3.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BRACE AXIAL STRESS CHECK: The axial stress in the diagonals, calculated using the Quick Check procedure of Section 4.4.3.4 is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel moment frames.	5.7.2	A.5.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the elastic moment (F_yS) of the adjoining members.	5.5.2.2.1	A.3.1.3.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COMPACT MEMBERS: All frame elements meet compact section requirements in accordance with AISC 360, Table B4.1a.	5.5.2.2.5	A.3.1.3.8	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1	

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Connections						
C	**NC**	**N/A**	**U**	ROOF PANELS: Where considered as diaphragm elements for lateral resistance, metal, plastic, or cementitious roof panels are positively attached to the roof framing to resist seismic forces.	5.7.5	A.5.5.1
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	WALL PANELS: Where considered as shear elements for lateral resistance, metal, fiberglass, or cementitious wall panels are positively attached to the framing and foundation to resist seismic forces.	5.7.5	A.5.5.2
☐	☐	☐	☐			

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